

Lesson 60 Practice: Answers and Rationales

JKSSB Computer Awareness

October 10, 2026

PYQ provenance: Q1 adapts JKSSB Junior Assistant 2026 Q70 (deadlock recovery by resource preemption) and Q3 adapts JA 2026 Q71 (priority inversion/inheritance). The cited key is the official final key for JA 2026, and both adaptations are paraphrased, not verbatim.

- Q1. A.** Resource preemption recovers from deadlock by forcibly taking a resource from one process and giving it to another, breaking the cycle. B, C, and D are necessary *conditions* for deadlock, not recovery methods.
- Q2. C.** The four necessary conditions are mutual exclusion, hold and wait, *no preemption*, and circular wait. “Preemption” is used to *recover* from deadlock, so it is not a condition. A, B, and D are genuine conditions.
- Q3. A.** A high-priority task waiting behind a low-priority holder that is itself preempted by a medium-priority task is priority inversion. B is the cure, not the problem; C requires a circular resource dependency; D concerns memory, not priorities.
- Q4. B.** Priority inheritance temporarily raises the holder to the waiting task’s priority so it releases the resource quickly. Time slicing (A), spooling (C), and paging (D) do not address priority inversion.
- Q5. C.** Round-robin gives each ready process a turn but ignores resource holding, so by itself it does not prevent priority inversion. Priority inheritance (A), the priority ceiling protocol (B), and priority boosting (D) all address inversion.
- Q6. A.** Being repeatedly passed over while others are served is starvation. Deadlock (B) is a circular wait with no progress possible; thrashing (C) is excessive paging; fragmentation (D) concerns memory or disk layout.
- Q7. A.** Aging gradually raises the priority of a long-waiting process so it eventually runs. Swapping (B), spooling (C), and caching (D) are unrelated storage techniques.
- Q8. B.** Preemptive priority scheduling interrupts the running process as soon as a higher-priority process becomes ready. A and C describe non-preemptive behaviour; D is not a preemption trigger.
- Q9. A.** RTOS expands to Real-Time Operating System, designed to meet time deadlines. B, C, and D are incorrect expansions.
- Q10. C.** Statements I and III are true. II is false: round-robin by itself does not prevent priority inversion. A includes false II; B includes II and omits I; D includes II.

- Q11. A.** The core that manages processes, memory, devices, and file systems is the Linux kernel. The shell (B) is the command interpreter; GNOME (C) is a desktop environment; GRUB (D) is a bootloader.
- Q12. A.** `/root` is the home directory of the superuser; `/home` holds ordinary users' home directories. B reverses the two; C mislabels `/root`; D mislabels `/home` as the top of the file system (which is `/`).
- Q13. B.** `ext4` is the native, common default file system for Linux. NTFS (A) is the Windows default; FAT32 (C) is used for broad compatibility; `exFAT` (D) is a removable-media file system, not the Linux native default.
- Q14. B.** NTFS is the default for modern Windows installations. `ext4` (A) and `ext3` (C) are Linux file systems; HFS+ (D) is an older Apple file system.
- Q15. C.** The Windows Store (Microsoft Store) is the built-in online marketplace for downloading and installing apps. It is not a compression tool (A), a defragmenter (B), or a Linux package manager (D).
- Q16. B.** Notepad is a plain-text editor, while WordPad supports basic rich-text formatting such as bold and fonts. A swaps them; C mislabels both as spreadsheets; D is false, since WordPad can save files (for example in `.rtf`).
- Q17. A.** Disk Cleanup frees disk space by removing temporary files and the Recycle Bin contents. The Defragmenter (B) rearranges file fragments; Device Manager (C) manages hardware; Task Scheduler (D) runs tasks on a schedule.
- Q18. A.** An SSD has no moving parts, so defragmenting gives no speed benefit and only adds write wear; Windows uses TRIM instead. B is irrelevant to defragmentation; C is false (an SSD is non-volatile); D is false.
- Q19. A.** Windows Fax and Scan is the imaging and fax utility used to scan documents and send or receive faxes. WordPad (B) is a text editor; Disk Cleanup (C) frees space; the Windows Store (D) installs apps.
- Q20. A.** `xcopy /s` copies the named directory including its subdirectories, excluding empty ones. It does not delete files (B), rename a file (C), or format a disk (D).
- Q21. A.** The `ls` command lists files. B, C, and D are not Linux commands.
- Q22. A.** Waiting processes are blocked. B, C, and D are not process states.
- Q23. A.** Linux is open source and multi-user. B, C, and D name unrelated objects.
- Q24. A, B, C.** `ls`, `cd`, and `mkdir` are standard utilities. D is not a command.
- Q25. A, B, C.** The process definition, scheduler role, and blocked state are correct. D is false: processes are runtime entities.
- Q26. A.** Resource preemption reclaims a resource to break deadlock. The other actions do not change process-resource allocation.
- Q27. A.** `'pwd'` prints the current directory. `'ls'` lists entries, `'cd'` changes directory, and `'rm'` removes files.

- Q28. A.** 'ls' lists directory entries. 'pwd' prints the path; 'mkdir' and 'rmdir' create/remove directories.
- Q29. A.** Round-robin cycles through ready processes using a time quantum. The other options do not specify that rotating time-slice policy.
- Q30. A, B, C.** Processes, deadlocks, and file systems are OS concepts. A driver is software that helps control a device.

Concepts That Click